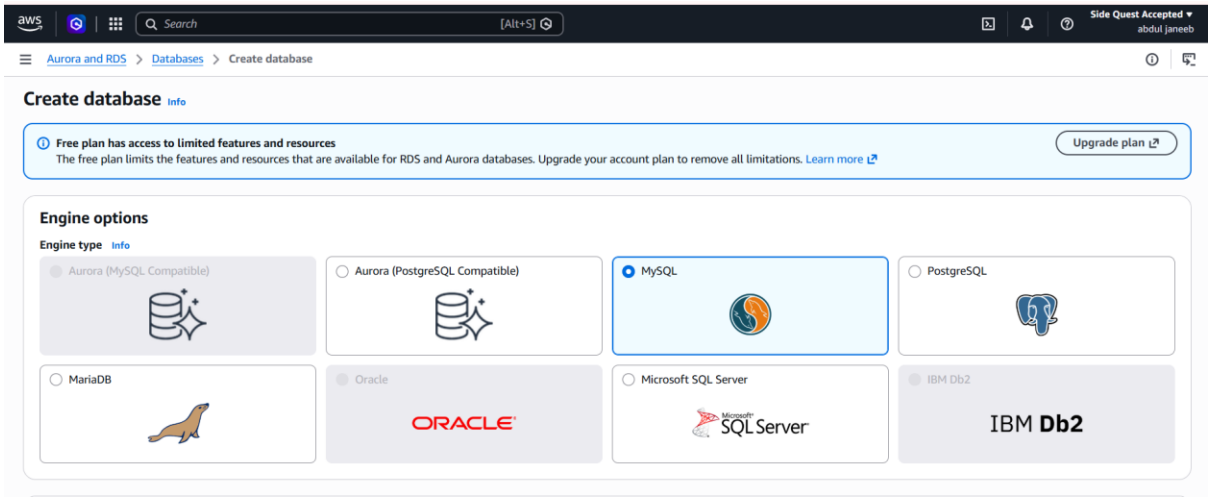
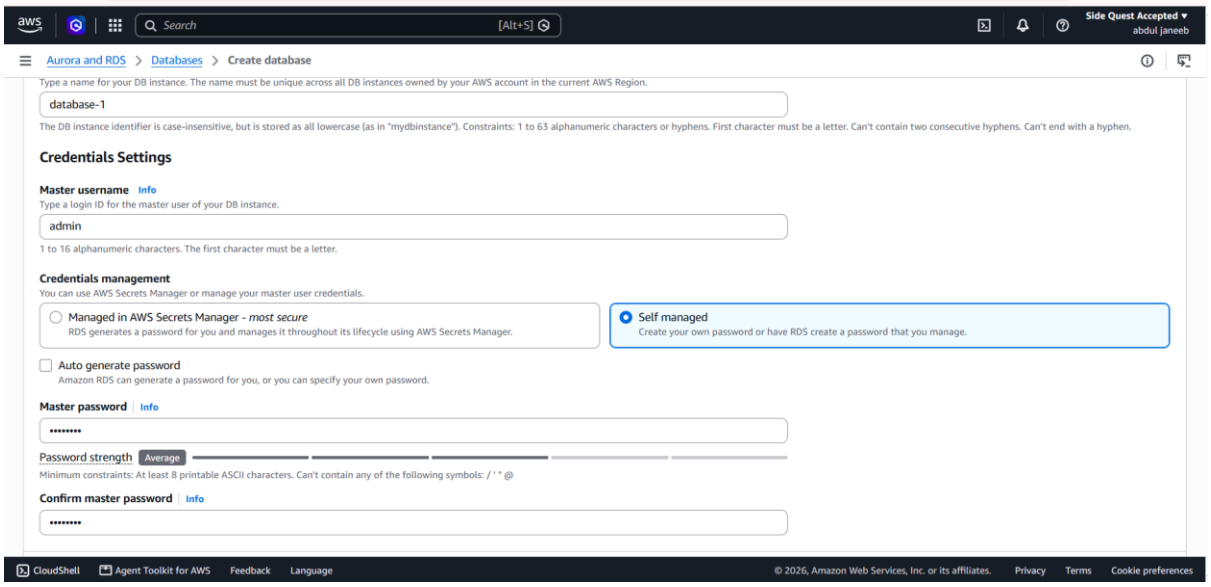
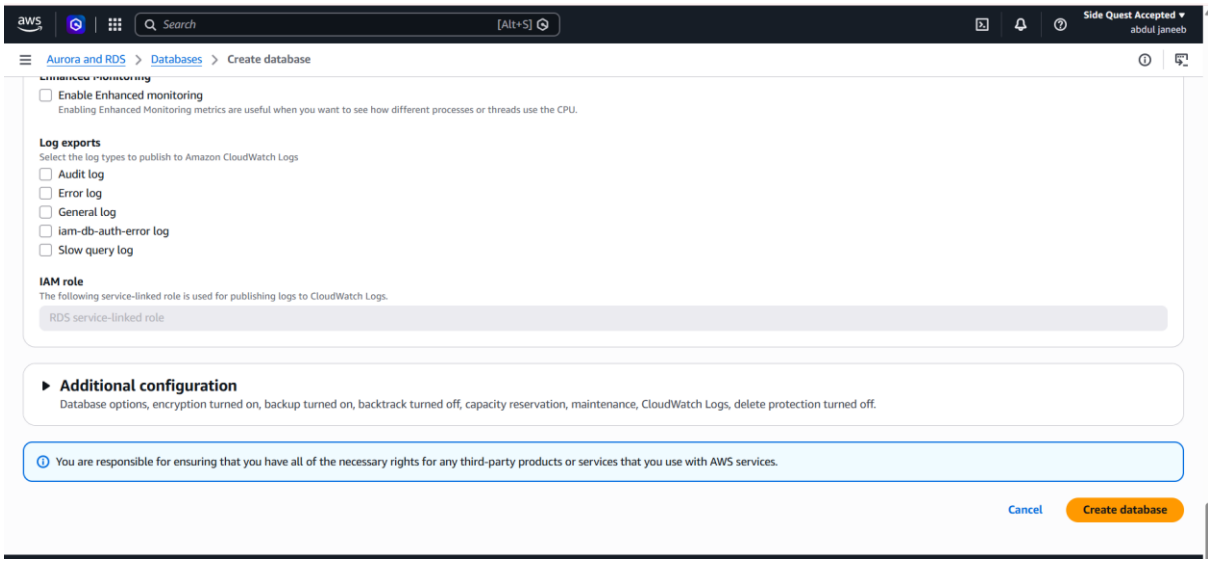
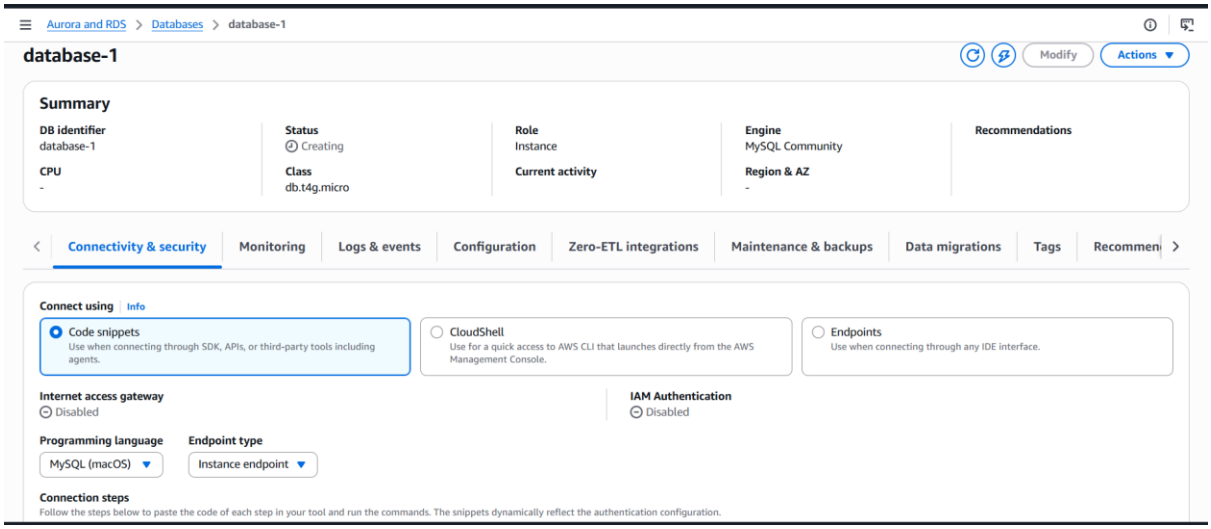
- Creating amazon RDR database



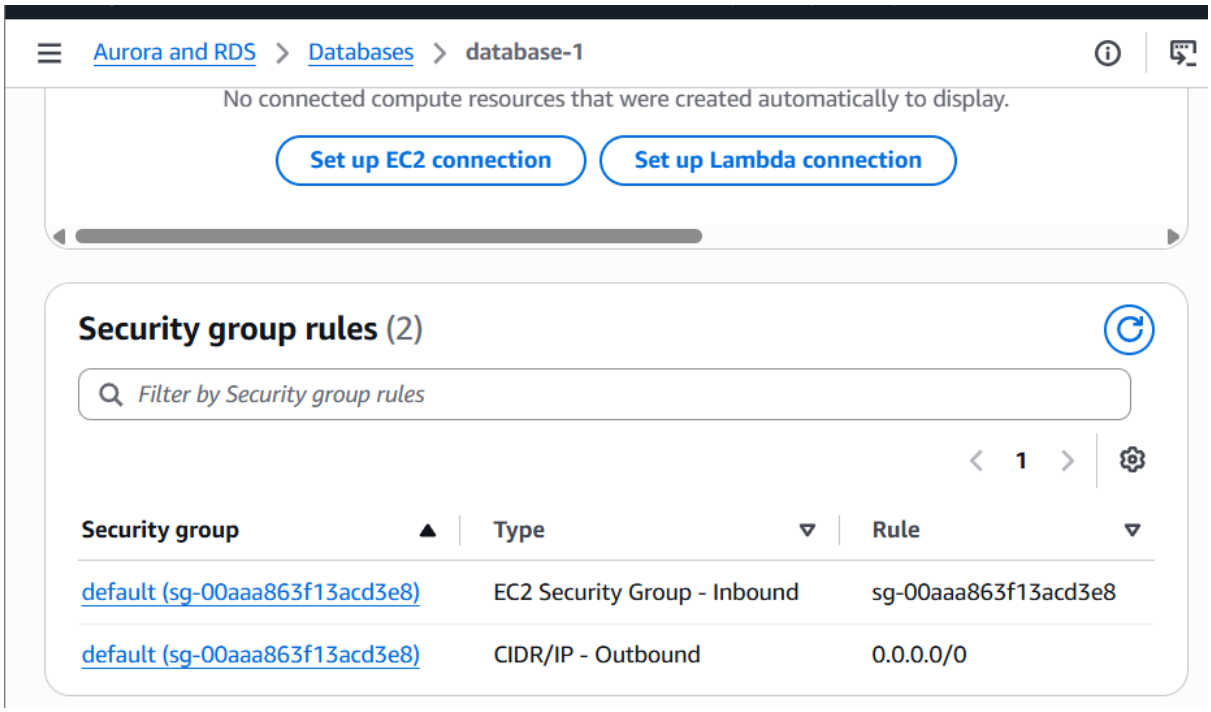




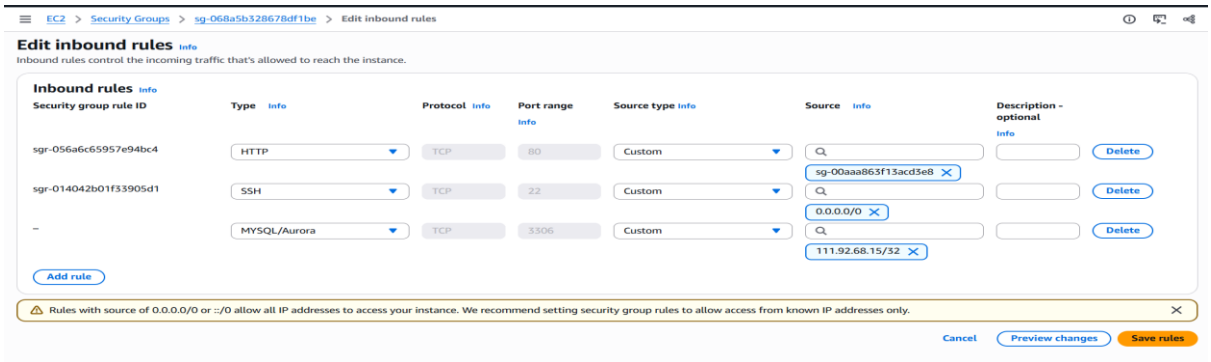
- Database is created successfully



- Configuring security group



- I added inbound security rule to RDS database



- Connecting from my system to DB

Command Prompt

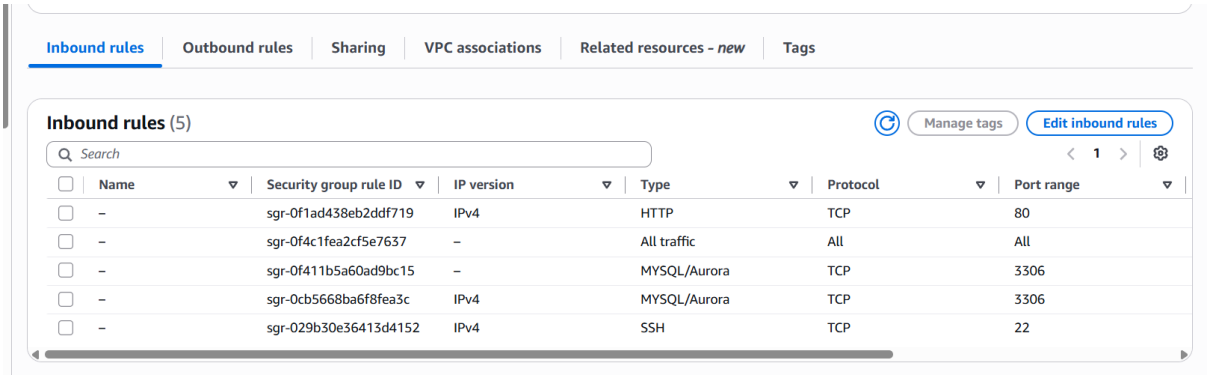
```
Microsoft Windows [Version 10.0.22000.2538]
(c) Microsoft Corporation. All rights reserved.

C:\Users\hp>curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  170k  100  170k    0     0   165k      0  0:00:01  0:00:01 --:--:--  166k

C:\Users\hp>mysql -h database-1.c504ogck2e9b.ap-southeast-2.rds.amazonaws.com -P 3306 -u admin -p --ssl-mode=
'mysql' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\hp>
```

- Error founded
- So I created a EC2 instance instead of from my system here I choose ec2
- After that I added inbound rules to ec2 instance such as



- Then I connected EC2 instance from system

```
C:\Users\hp\Documents\awskey>ssh -i "ec2-web-key.pem" ubuntu@ec2-13-210-104-33.ap-southeast-2.compute.amazonaws.com
The authenticity of host 'ec2-13-210-104-33.ap-southeast-2.compute.amazonaws.com (13.210.104.33)' can't be established.
ECDSA key fingerprint is SHA256:XtL8YtjxKJ/l+8mKeo8FhIHJtnkHtFzziQB/LN9FONI.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-13-210-104-33.ap-southeast-2.compute.amazonaws.com,13.210.104.33' (ECDSA) to the list of known hosts.
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-1014-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Nov  2 10:36:17 UTC 2026

  System load:  0.08              Temperature:           -273.1 C
  Usage of /:   30.4% of 6.61GB   Processes:             116
  Memory usage: 25%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.47.82

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

- Installed mysql

```
31 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-47-82:~$ sudo apt install mysql-client -y
Installing:
  mysql-client

Installing dependencies:
  libgoogle-perftools4t64  libtcmalloc-minimal4t64  mysql-client-core  mysql-common

Summary:
  Upgrading: 0, Installing: 5, Removing: 0, Not Upgrading: 31
  Download size: 2365 kB
  Space needed: 49.5 MB / 4680 MB available

Get:1 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libtcmalloc-m
Get:2 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 libgoogle-per
Get:3 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 mysql
Get:4 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 mysql-common
Get:5 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 mysql
Fetched 2365 kB in 0s (9315 kB/s)
Selecting previously unselected package libtcmalloc-minimal4t64:amd64.
(Reading database ... 85362 files and directories currently installed.)
Preparing to unpack .../libtcmalloc-minimal4t64_2.18.1-1_amd64v3.deb ...
Unpacking libtcmalloc-minimal4t64:amd64 (2.18.1-1) ...
Selecting previously unselected package libgoogle-perftools4t64:amd64.
Preparing to unpack .../libgoogle-perftools4t64_2.18.1-1_amd64v3.deb ...
Unpacking libgoogle-perftools4t64:amd64 (2.18.1-1) ...
Selecting previously unselected package mysql-client-core.
Preparing to unpack .../mysql-client-core_8.4.11-0ubuntu0.26.04.1_amd64v3.deb ...
Unpacking mysql-client-core (8.4.11-0ubuntu0.26.04.1) ...
Selecting previously unselected package mysql-common.
Preparing to unpack .../mysql-common_5.8+1.1.1ubuntu2_all.deb ...
```

- After that connected to RDS database

```
mysql  Ver 8.4.11-0ubuntu0.26.04.1 for Linux on x86_64 ((Ubuntu))
ubuntu@ip-172-31-47-82:~$ curl -o global-bundle.pem https://truststore.pki.rds.amazonaws.com/global/global-bundle.pem
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 170.4k 100 170.4k    0     0 225.4k      0                              0
ubuntu@ip-172-31-47-82:~$ mysql -h database-1.c504ogck2e9b.ap-southeast-2.rds.amazonaws.com -P 3306 -u admin -p --ssl-mode=VERIFY_IDENTITY --ssl-ca=./global-bundle.pem
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 64
Server version: 8.4.11 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.02 sec)

mysql> SELECT VERSION();
+-----------+
| VERSION() |
+-----------+
| 8.4.11    |
+-----------+
1 row in set (0.00 sec)

mysql> SELECT DATABASE();
+------------+
| DATABASE() |
+------------+
| NULL       |
+------------+
1 row in set (0.01 sec)
```

- Created a database using mysql (company)

```
ubuntu@ip-172-31-47-82:~$ mysql -h database-1.c504ogck2e9b.ap-southeast-2.rds.amazonaws.com -P 3306 -u admin -p --ssl-mode=VERIFY_IDENTITY --ssl-ca=./global-bundle.pem
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 125
Server version: 8.4.11 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| company            |
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.00 sec)
```

- Created a table called employees inserted 3 users

```
mysql> USE company;
Database changed
mysql> CREATE TABLE employees (id INT PRIMARY KEY,name VARCHAR(50),department VARCHAR(50),salary INT);
Query OK, 0 rows affected (0.05 sec)

mysql> SHOW TABLES;
+-------------------+
| Tables_in_company |
+-------------------+
| employees         |
+-------------------+
1 row in set (0.00 sec)

mysql> INSERT INTO employees VALUES(1,'janeeb','IT',30000),(2,'fazil','HR',40000),(3,'Noufal','Finance','500);
    '> \c
    '> ^C
mysql> select * from employees
    -> ^C
mysql> select * from employees;
Empty set (0.02 sec)

mysql> USE company;
Database changed
mysql> insert into employees (id, name, department, salary)values(1, 'janeeb', 'IT', 30000),(2, 'fazil', 'HR', 30000),(3, 'noufal', 'DEV', 20000);
Query OK, 3 rows affected (0.01 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql> select * from employees;
+----+--------+------------+--------+
| id | name   | department | salary |
+----+--------+------------+--------+
|  1 | janeeb | IT         |  30000 |
|  2 | fazil  | HR         |  30000 |
|  3 | noufal | DEV        |  20000 |
+----+--------+------------+--------+
3 rows in set (0.00 sec)
```

- Modified employee janeeb salary

```
mysql> update employees set salary = 45000 where id = 1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

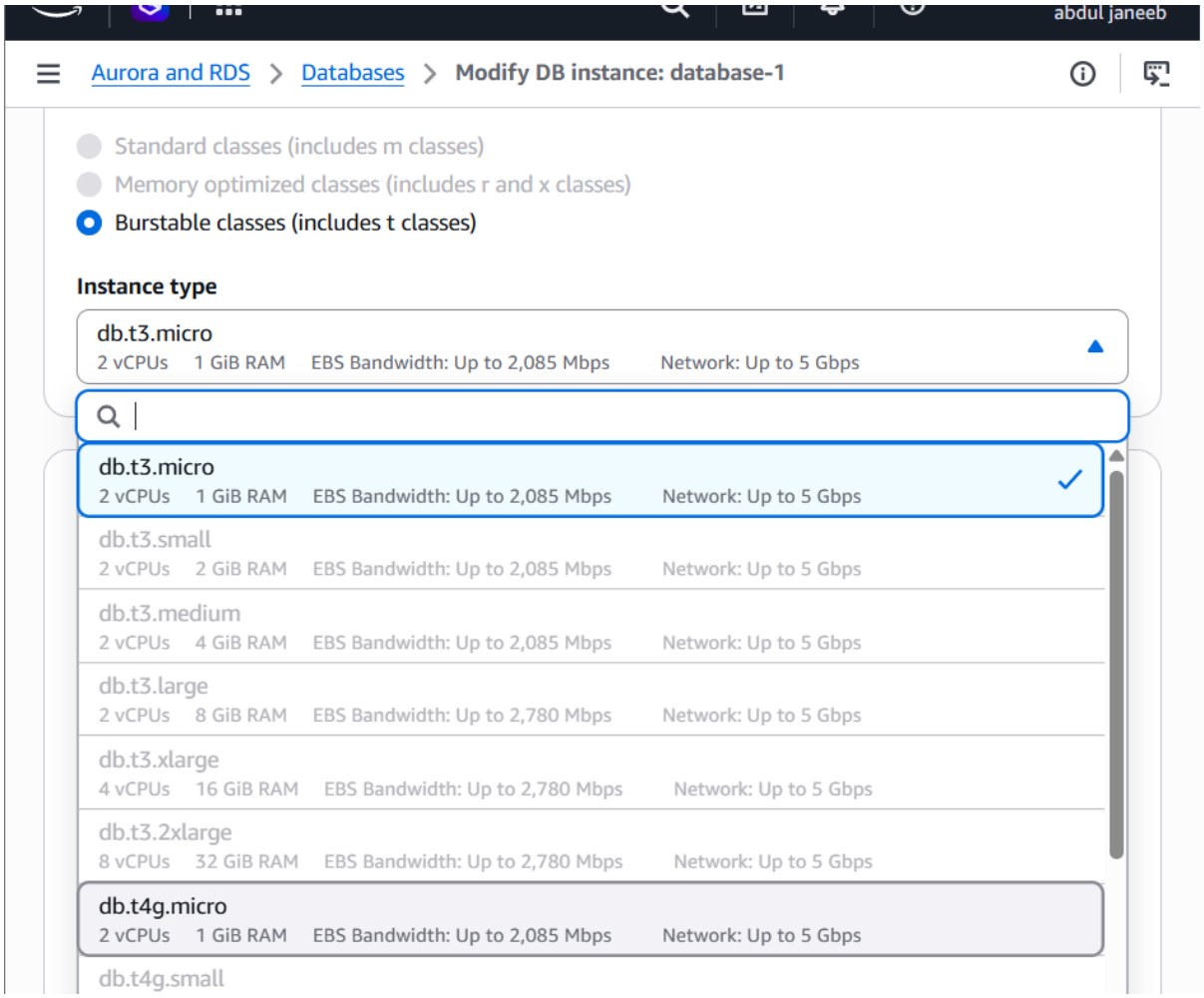
mysql> select * from employees;
+----+--------+------------+--------+
| id | name   | department | salary |
+----+--------+------------+--------+
|  1 | janeeb | IT         |  45000 |
|  2 | fazil  | HR         |  30000 |
|  3 | noufal | DEV        |  20000 |
+----+--------+------------+--------+
3 rows in set (0.00 sec)
```

- Deleted employee noufal from the table

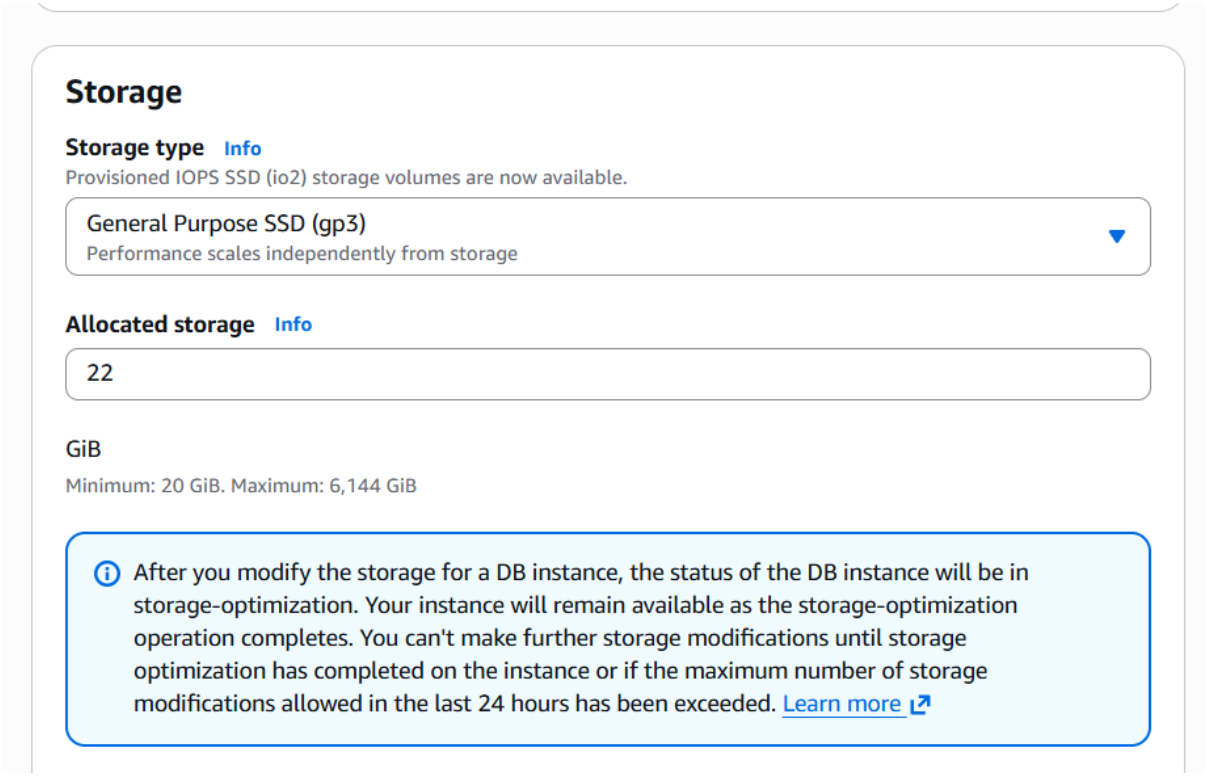
```
mysql> delete from employees where id = 3;
Query OK, 1 row affected (0.01 sec)

mysql> select * from employees
    -> ^C
mysql> select * from employees;
+----+--------+------------+--------+
| id | name   | department | salary |
+----+--------+------------+--------+
|  1 | janeeb | IT         |  45000 |
|  2 | fazil  | HR         |  30000 |
+----+--------+------------+--------+
2 rows in set (0.00 sec)
```

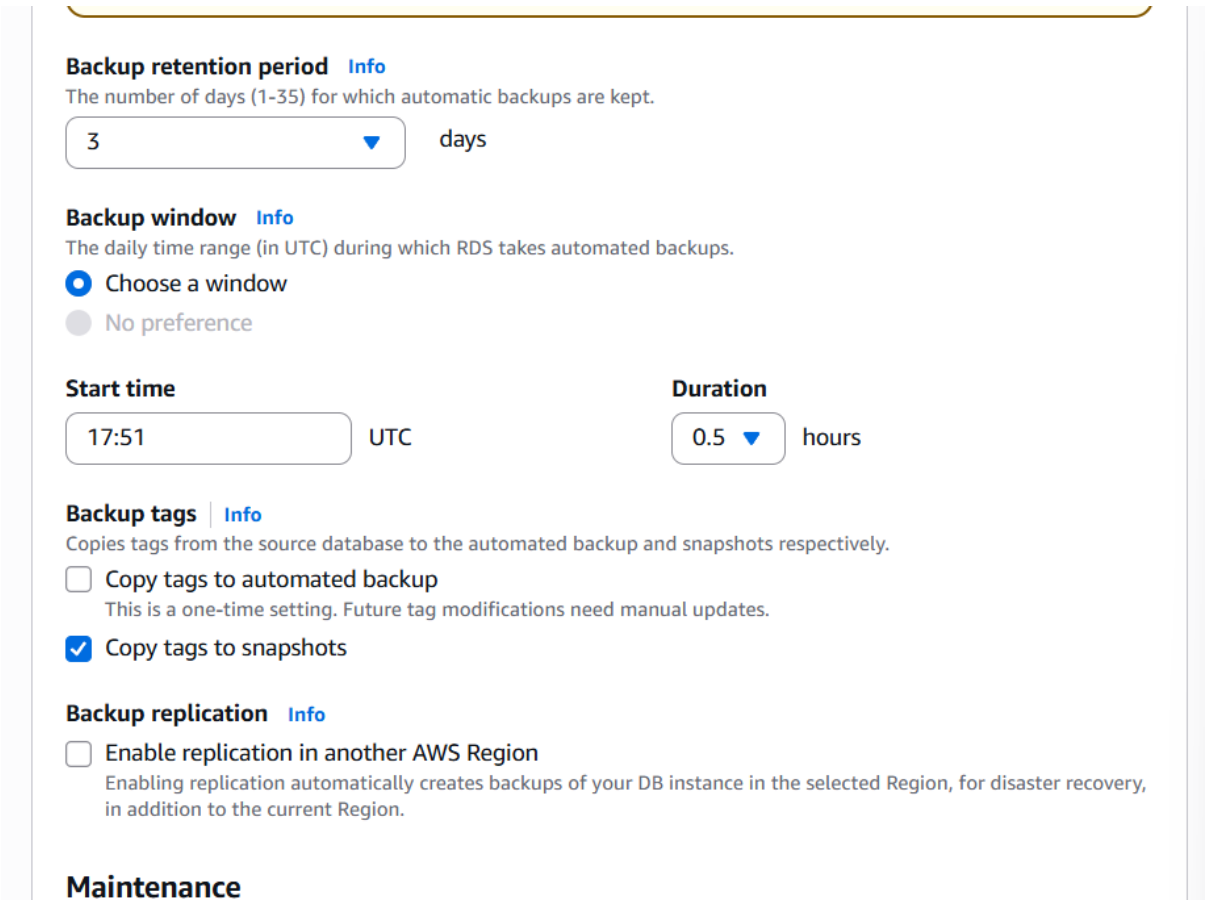
- Modified instance type



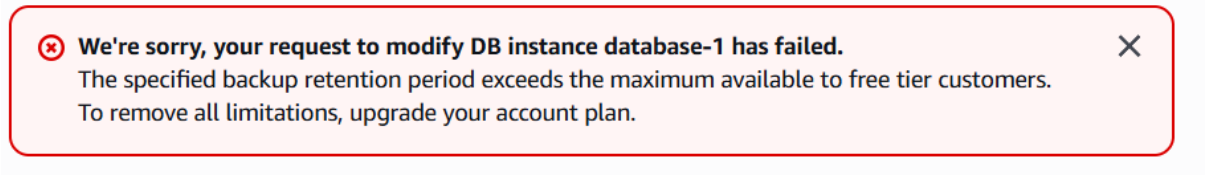
- Modified storage



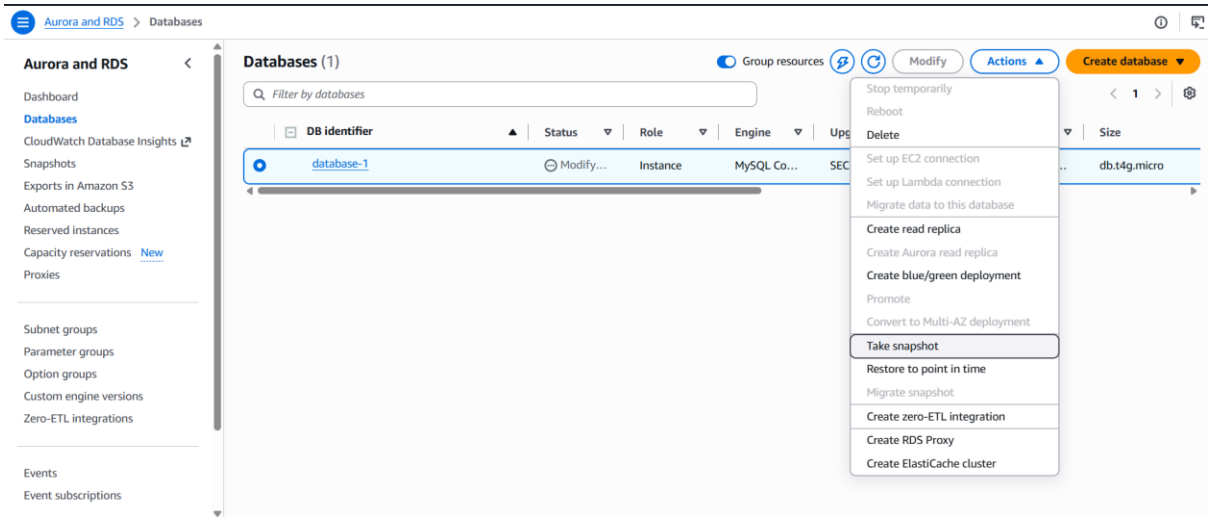
- Also modified backup retention period and window

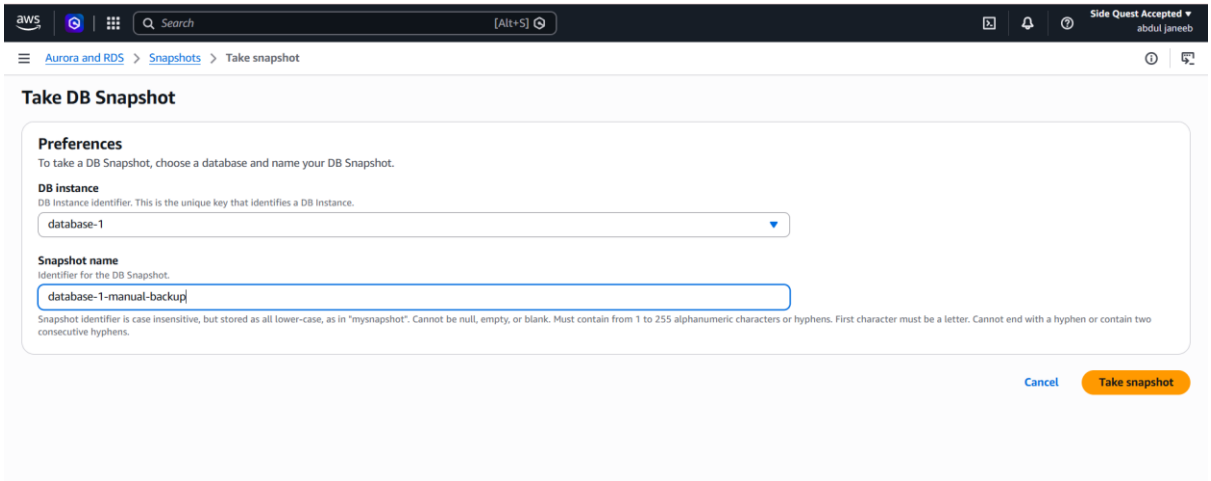


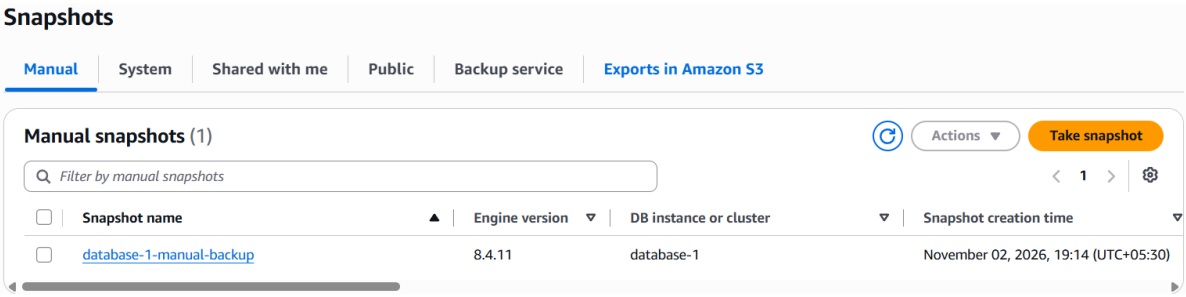
- Not possible for free tier increasing retention period



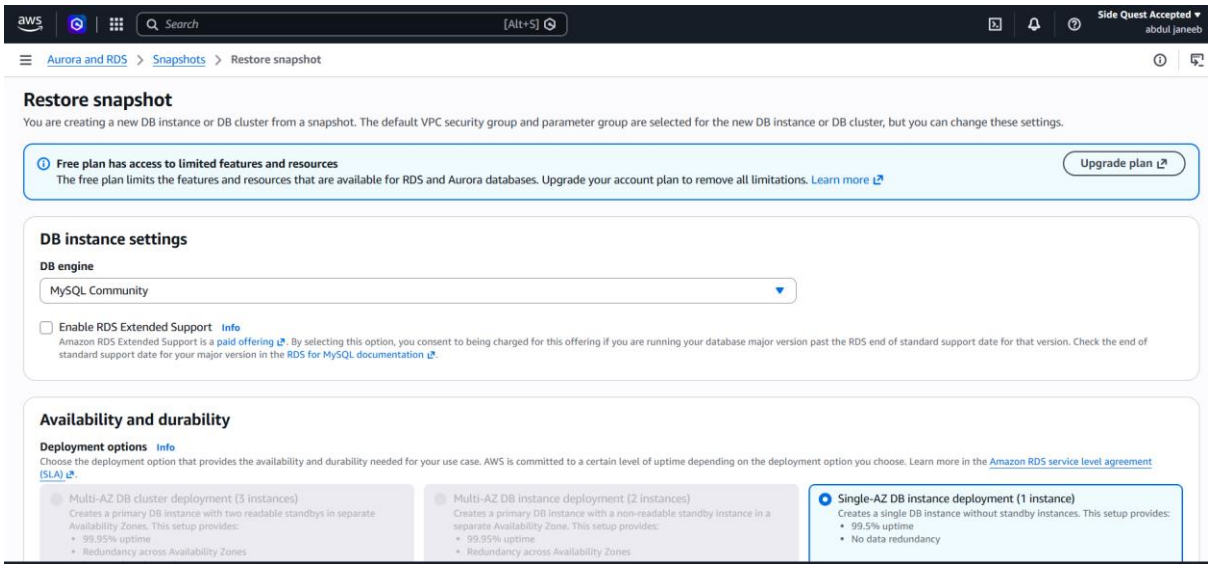
- Created a manual backup using snapshot



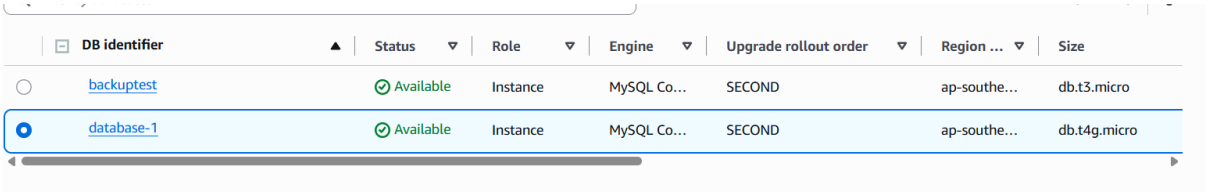




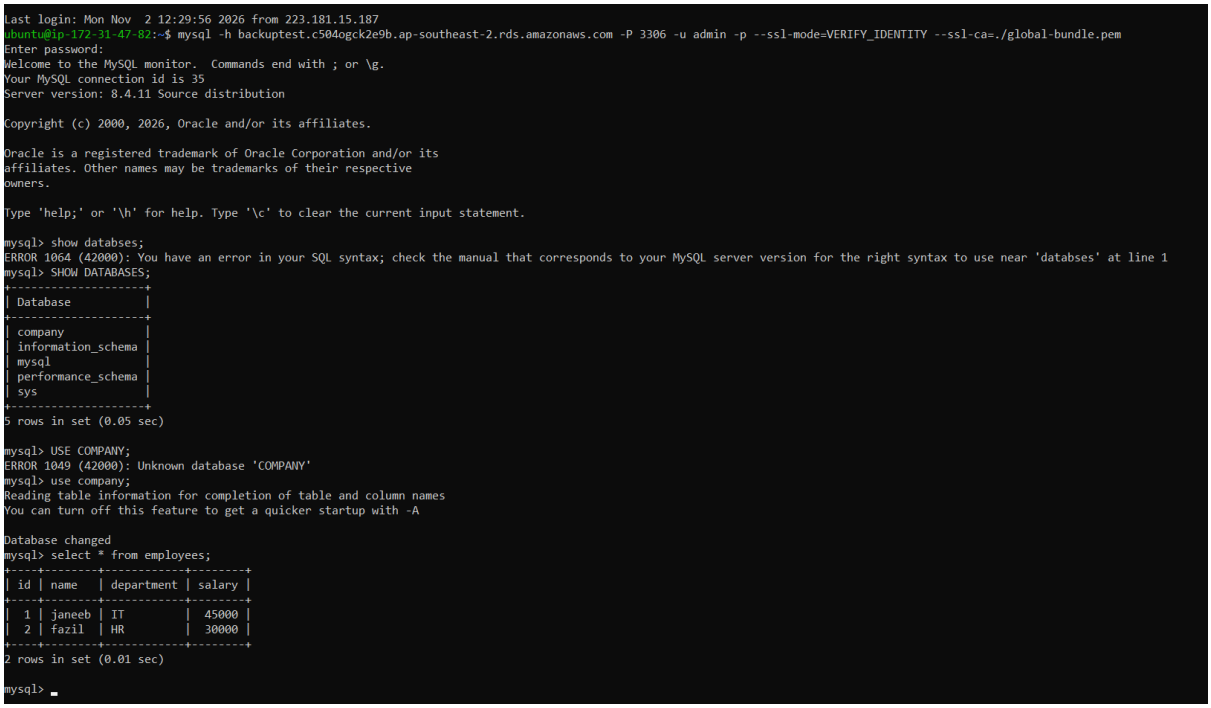
- After that I restored the backup snapshot



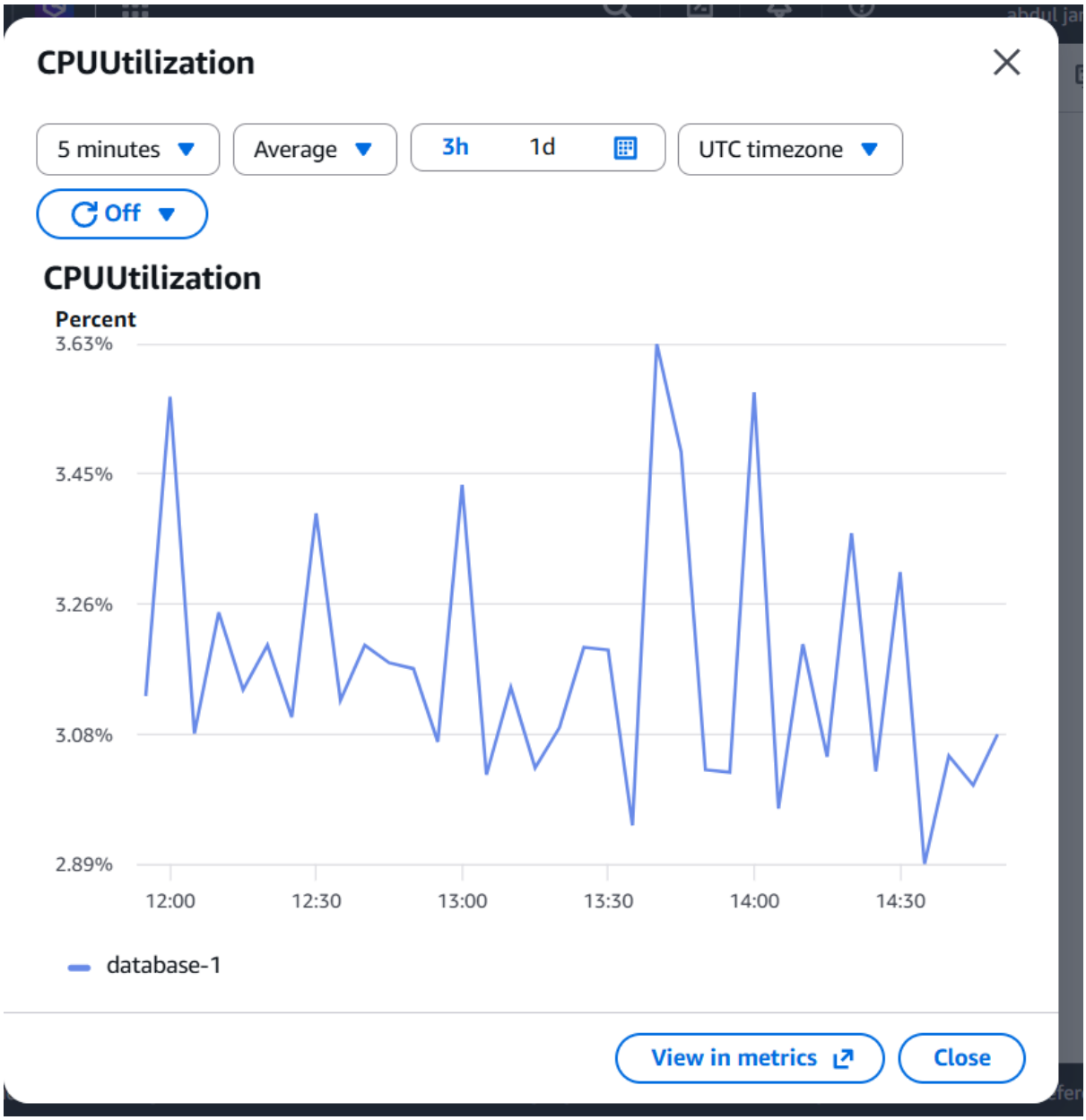
- Into new database because avoid overwriting .
- Restored database is backuptest



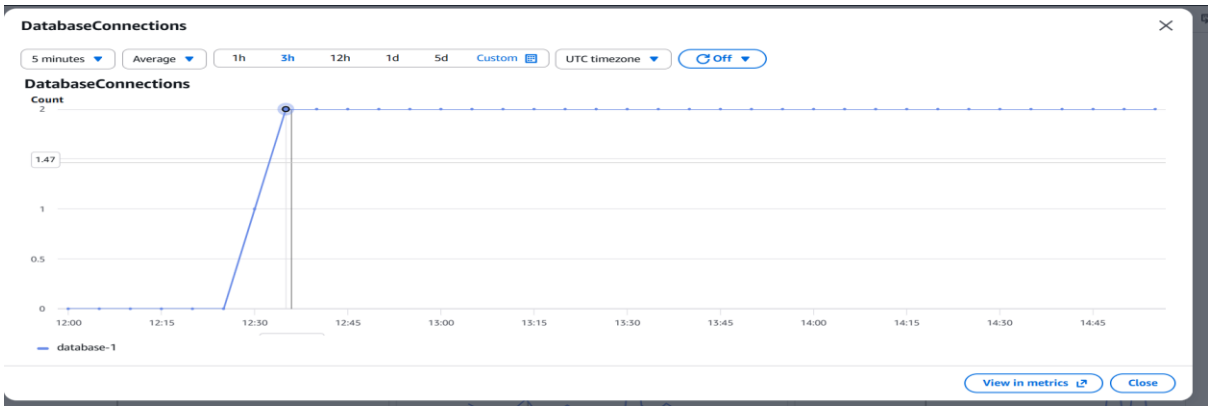
- After that verifying the database and the verification success



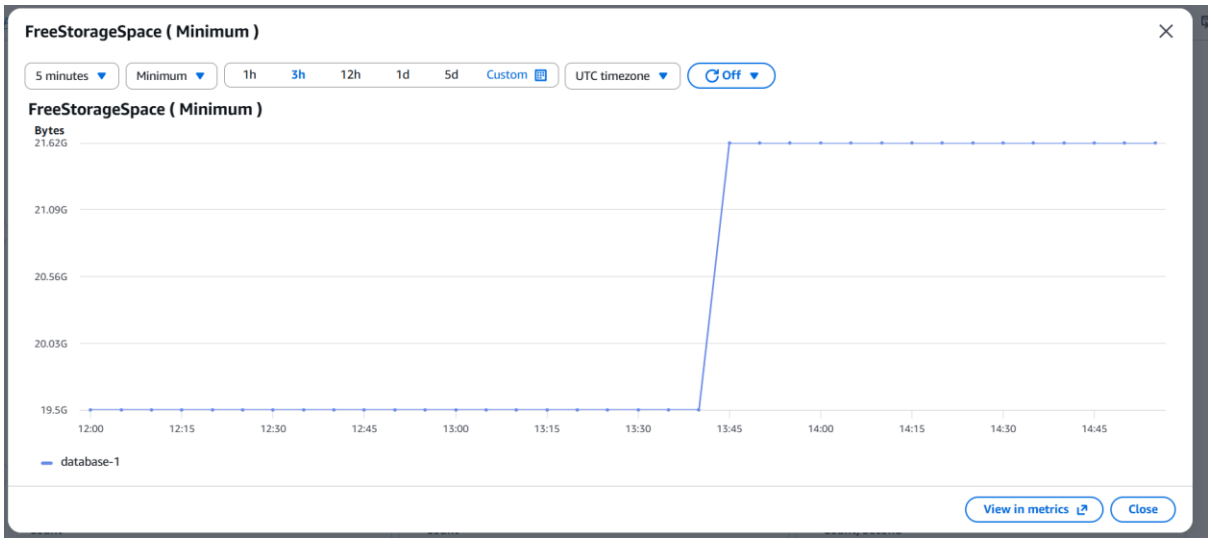
- Cloudwatch monitoring
- Cpu utilization



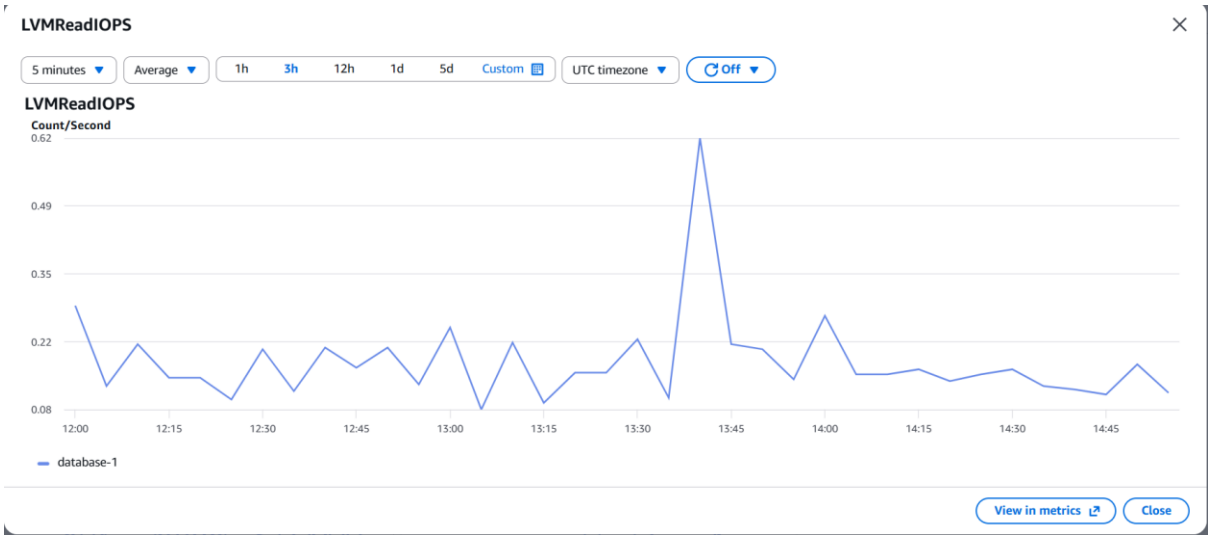
- DatabaseConnections



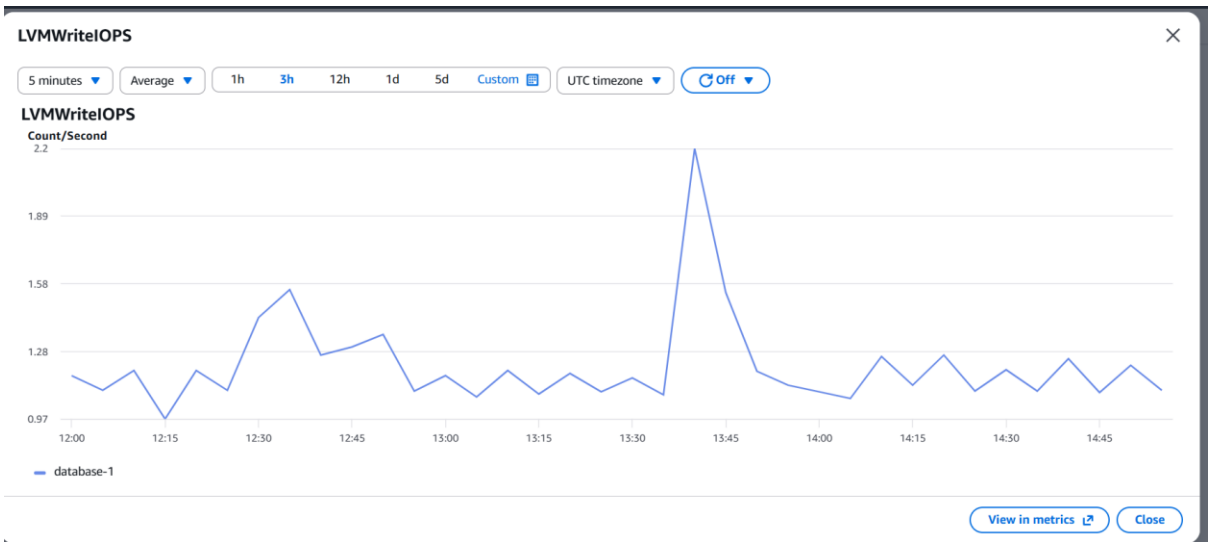
- FreeStorageSpace



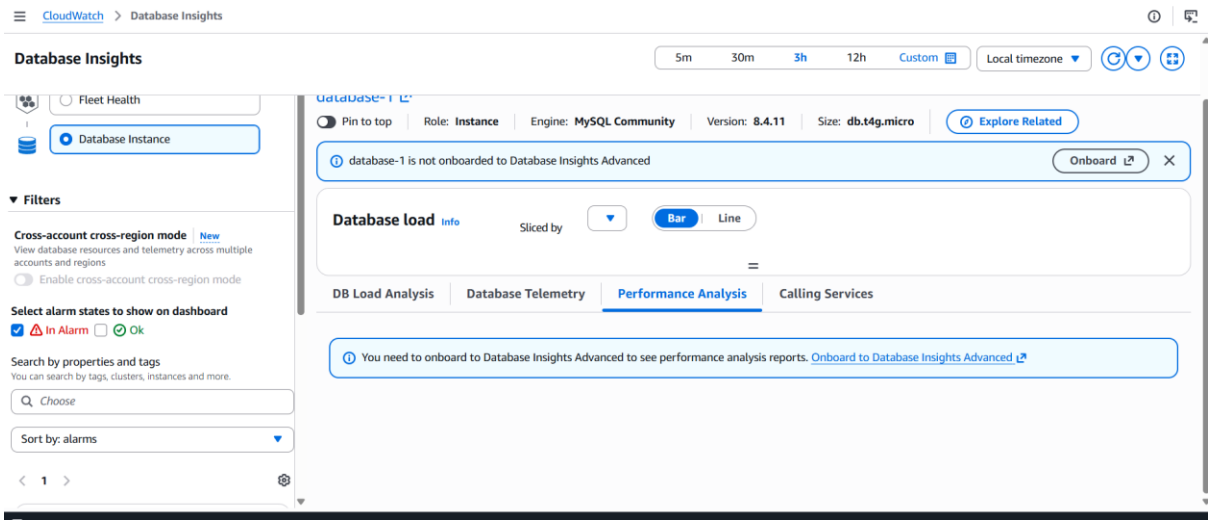
- ReadIOPS



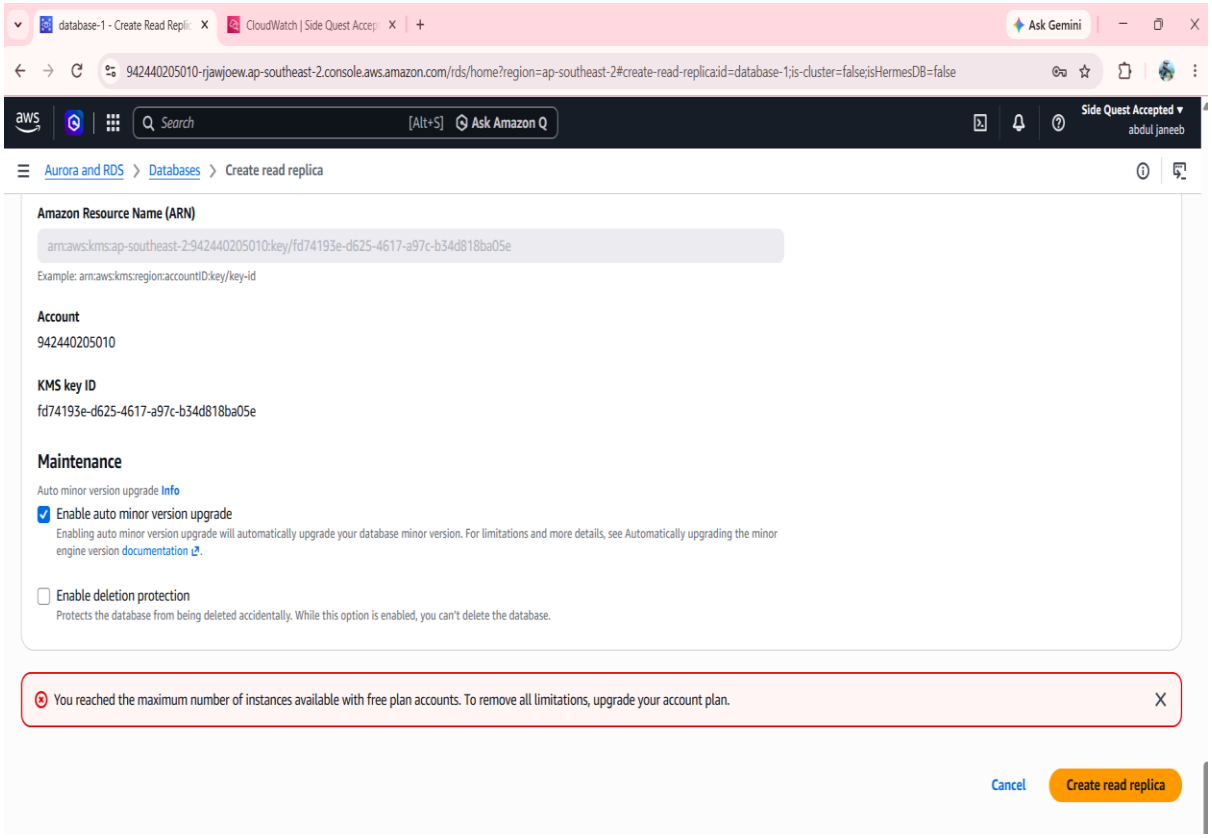
- WriteIOPS



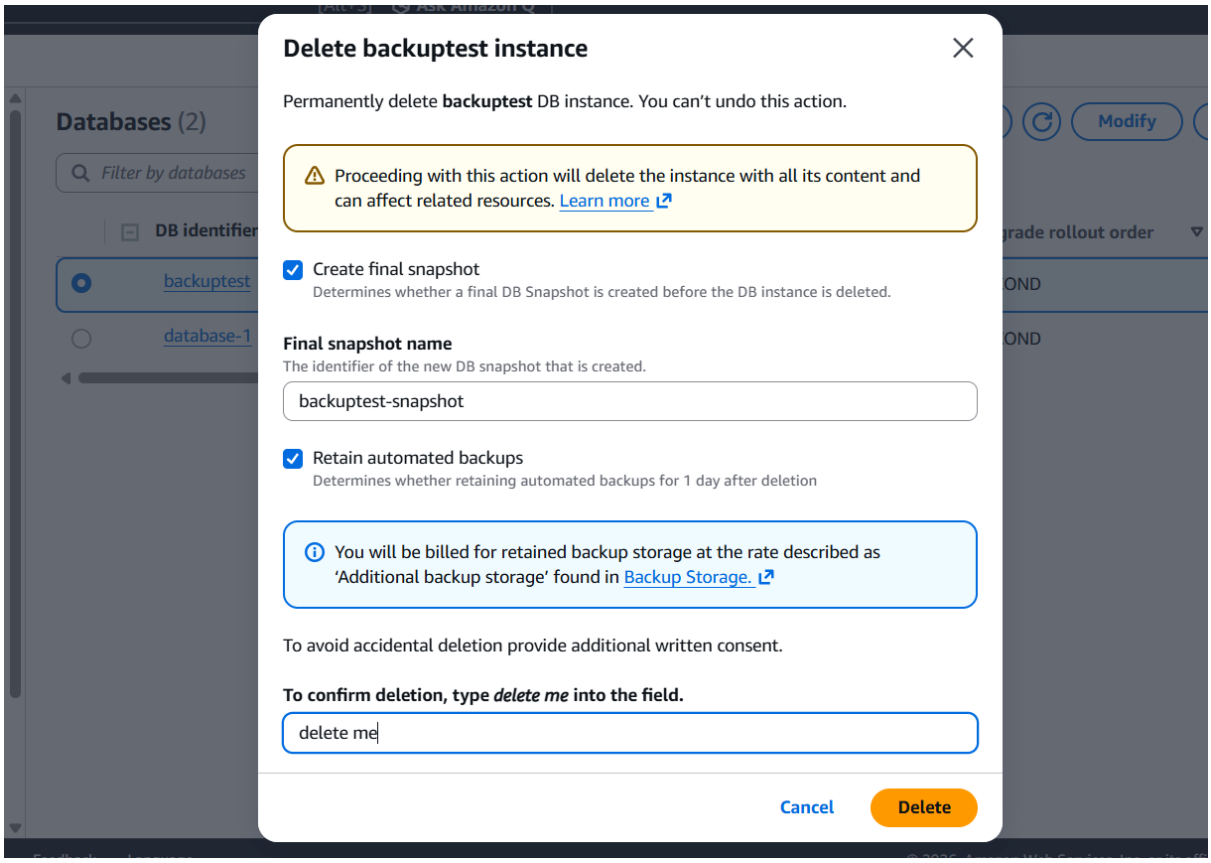
- Performance Insights was explored through AWS Database Insights. However, Performance Analysis reports required Database Insights Advanced onboarding, which was unavailable under the current Free Plan. Therefore, the limitation was documented without upgrading the account.

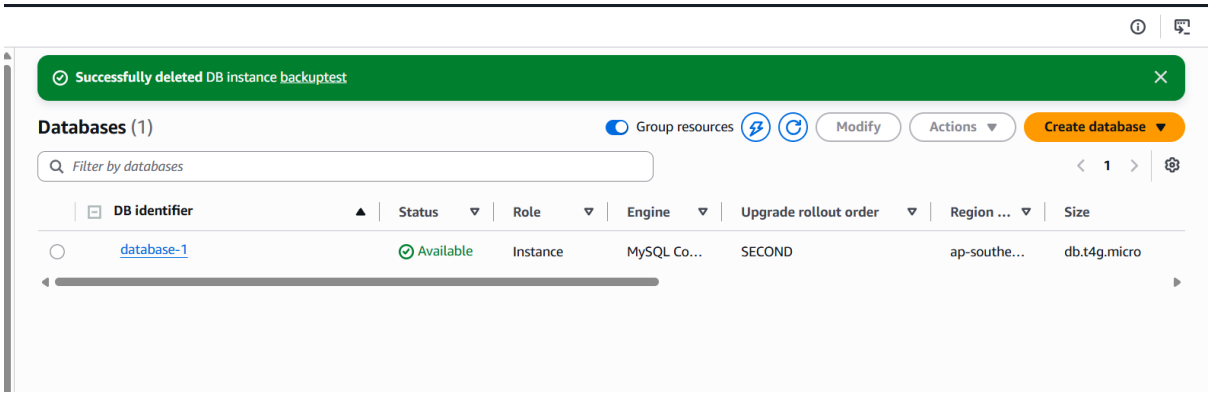


- Read Replica(optional)
- Attempted to create an RDS Read Replica, but AWS Free Plan restrictions prevented creation because the maximum number of available DB instances had been reached.



- Cleanup


Delete backuptest instance
Permanently delete backuptest DB instance. You can't undo this action.
Proceeding with this action will delete the instance with all its content and can affect related resources. Learn more
Create final snapshot
Determines whether a final DB Snapshot is created before the DB instance is deleted.
Final snapshot name
The identifier of the new DB snapshot that is created.
backuptest-snapshot
Retain automated backups
Determines whether retaining automated backups for 1 day after deletion
You will be billed for retained backup storage at the rate described as 'Additional backup storage' found in Backup Storage.
To avoid accidental deletion provide additional written consent.
To confirm deletion, type delete me into the field.
delete me
Cancel
Delete
Databases (2)
Filter by databases
DB identifier
backuptest
database-1
Modify


Successfully deleted DB instance backuptest
Databases (1)
Filter by databases
Group resources
Modify
Actions
Create database
DB identifier
Status
Role
Engine
Upgrade rollout order
Region ...
Size
database-1
Available
Instance
MySQL Co...
SECOND
ap-southe...
db.t4g.micro